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$$\begin{aligned}\int \sqrt{2x} dx &= \int (2x)^{\frac{1}{2}} dx \\&= \int 2^{\frac{1}{2}} \cdot x^{\frac{1}{2}} dx \\&= 2^{\frac{1}{2}} \cdot \int x^{\frac{1}{2}} dx \\&= \sqrt{2} \int x^{\frac{1}{2}} dx \\&= \sqrt{2} \cdot \frac{x^{\frac{3}{2}}}{\frac{3}{2}} \\&= \frac{2\sqrt{2} \cdot \sqrt{x^3}}{3} = \frac{2x\sqrt{2x}}{3} + C\end{aligned}$$

References